

Technology Stack

Date	25-06-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, HTML, CSS	Provides a simple, interactive, and responsive web interface where students and educators can enter learning problems and view AI-generated emotion analysis and personalized guidance.
2	Backend / Server-Side	Python 3.11, TensorFlow, Keras, Hugging Face Transformers	Python serves as the core programming language. TensorFlow/Keras is used for the BiLSTM model, while Hugging Face Transformers is used for the BERT emotion detection model.

3	Database / Data Storage	CSV Files, Local File Storage (Optional: PostgreSQL/Firebase)	Stores user interaction history, detected emotions, confidence scores, timestamps, and AI responses for future analysis and reporting.
4	Cloud / Hosting / Deployment	Google Cloud Run, Docker, Google Artifact Registry	Deploys the application as a scalable cloud service with automatic scaling, secure container management, and reliable hosting.
5	Version Control & CI/CD	Git, GitHub	Enables collaborative development, version tracking, source code management, and easy project deployment by multiple team members.
6	Third-Party APIs / Other Tools	Google Gemini API, Plotly, NLTK, Kaggle GPU	Gemini API generates personalized learning responses. Plotly creates interactive analytics dashboards. NLTK performs text preprocessing. Kaggle GPU is used to train the BiLSTM and BERT models efficiently.